

Solution Requirements

Date	20 June 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users (Farmers, Researchers, Admin) can securely log in using a username and password.	High
2	Authorization Levels	Admin manages users and data, while farmers and researchers can access crop recommendations and reports based on their roles.	High
3	External Interfaces	The system integrates with crop datasets, weather data sources, and the Machine Learning prediction model.	High
4	Transaction Processing	The system accepts soil and environmental inputs (N, P, K, pH, temperature, humidity, rainfall), processes them, and generates crop recommendations.	High
5	Reporting	Displays crop recommendations, crop suitability, environmental analysis, and prediction results	Medium

		through reports and dashboards.	
6	Business Rules	Crop recommendations are generated only after validating all required soil and climate parameters using the ML model.	High
7	Compliance to Laws or Regulations	User data is stored securely and follows applicable data privacy and security standards.	Medium
8	Other	Allows users to view previous recommendations and improve future farming decisions.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should generate crop recommendations quickly.	Results displayed within 2–3 seconds .
2	Scalability	The system should support multiple users simultaneously.	Supports 500+ concurrent users .
3	Security & Data Privacy	User credentials and agricultural data should be securely stored and protected.	Password encryption and secure database access.
4	Reliability & Availability	The system should be available whenever users need it.	99% uptime with reliable prediction service.
5	Usability & Accessibility	The application should have a simple, user-friendly interface for farmers.	Easy navigation with responsive design.
6	Other	The system should be maintainable and allow future updates for new crops and prediction models.	Easy maintenance and extensibility.